23A01604b • OPEN CHANNEL FLOW

UNIT – 3

Unsteady Flows in Open Channels

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Equations of Motion (Saint-Venant equations)
- Uniformly Progressive Wave
- Rapidly Varied Unsteady Flow – Positive and Negative Surges
- Dam Break Problem

1. INTRODUCTION TO UNSTEADY OPEN-CHANNEL FLOW

1.1 What is unsteady flow?

Unsteady flow is flow in which depth and velocity change with time at a fixed location, i.e. y = y(x, t) and V = V(x, t). It is always non-uniform, because a change in time implies a change in space as the disturbance travels along the channel. Steady flow is only an idealisation; real channels carry flood waves, tidal bores, surges from gate operation, and dam-break waves.

1.2 Common examples

- **Flood waves** in rivers (slow, long waves caused by rainfall–runoff).
- **Surges** produced by sudden opening or closing of a gate, turbine load rejection in a power canal, or tidal bores in estuaries.
- **Dam-break waves** – extremely rapid and destructive.
- **Tides and seiches** in estuaries and lakes; **irrigation canal operation** (gate adjustments).

1.3 Classification of unsteady flow

Type	Feature	Example
Gradually varied unsteady	Long wave; vertical acceleration negligible; hydrostatic pressure	Flood routing
Rapidly varied unsteady	Abrupt change in depth; steep wave front; surface curvature important	Surges, bores, dam break

1.4 Wave celerity (small-amplitude long wave)

A small disturbance in still water of depth y moves with the **celerity**

$$\mathbf{c = \sqrt{g\,y}\ (\text{rectangular channel}) \text{ or } c = \sqrt{g\,A/T} = \sqrt{g\,D}}$$

Relative to a moving stream of velocity V: absolute wave speeds are **V ± c**. If Fr = V/c < 1, a wave can travel upstream; if Fr > 1, all waves are swept downstream. This is why Fr is the key parameter in open-channel dynamics.

1.5 Methods of analysis

- **Saint-Venant equations** with characteristics or finite-difference methods (full dynamic wave).
- **Approximate routing** (kinematic wave, diffusion wave, Muskingum).
- **Surge theory** (momentum + continuity across the wave front).

2. EQUATIONS OF MOTION – THE SAINT-VENANT EQUATIONS

2.1 Assumptions

- Flow is one-dimensional: depth and velocity vary only along the channel axis x.
- Streamline curvature is small; vertical acceleration is negligible, therefore pressure is **hydrostatic**.
- Channel bed slope is small ($\cos\theta \approx 1$); channel is straight with fixed bed (no scour or deposition).
- Boundary friction is represented by steady-flow formulas (Manning or Chezy) using the instantaneous V and R.
- The fluid is incompressible and of constant density.

2.2 Continuity equation

Consider an elementary reach dx. Mass in – mass out = increase of storage. Let lateral inflow per unit length be q_L.

$$\partial Q/\partial x + \partial A/\partial t = q_L$$

In terms of V and y for a rectangular/general channel (T = dA/dy): $\partial y/\partial t + V\,\partial y/\partial x + D\,\partial V/\partial x = 0$ (for $q_L = 0$, D = A/T).

2.3 Momentum (dynamic) equation

Newton's second law applied to the CV between x and x + dx. Forces: hydrostatic pressure force difference, gravity component, boundary friction. Result:

$$(1/g)\,\partial V/\partial t + (V/g)\,\partial V/\partial x + \partial y/\partial x = S_0 - S_f$$

Every term has the dimension of slope (dimensionless). They are called:

	Name	Meaning		
	Local acceleration term	Change of velocity with time at a section		
	Convective acceleration term	Change of velocity along x due to non-uniformity		
	Pressure (hydrostatic) term	Water surface gradient		
	Gravity (bed slope) term	Weight component along channel		
	Friction slope term	Resistance, $S_f = n^2V$	V	$/R^{4/3}$

2.4 The conservation form

Dependent variables A and Q:

$$\partial A/\partial t + \partial Q/\partial x = 0$$

$$\partial Q/\partial t + \partial(\beta Q^2/A)/\partial x + g\,A\,\partial y/\partial x = g\,A\,(S_0 - S_f)$$

This form is better for numerical schemes (finite-volume, Preissmann) and for dealing with shocks and bores.

2.5 Special cases – hierarchy of wave models

By dropping terms from the momentum equation we get simplified models:

Model	Terms retained	Equation	Use
Kinematic wave	$S_0 = S_f$	Q = Q(A) (uniform flow relation)	Steep slopes, overland flow, no backwater
Diffusion wave	$\partial y/\partial x$, S_0, S_f	Neglect local and convective acceleration	Flood routing with backwater, mild slopes
Quasi-steady dynamic wave	Convective + pressure + S_0, S_f	Neglect local acceleration	Slowly changing flows
Full dynamic wave	All terms	Complete Saint-Venant	Dam break, tides, surges, steep fronts

★ **Key Point:** Always start from the full equations and then identify which terms are negligible for the problem. Kinematic wave cannot reproduce attenuation or backwater; diffusion wave can.

2.6 Solution of the Saint-Venant equations

- **Method of characteristics:** the two PDEs are converted into four ODEs along the characteristic directions dx/dt = V ± c.
- **Explicit finite-difference schemes** (Lax diffusive, MacCormack, Leap-frog) need Courant condition $\Delta t \leq \Delta x/(|V| + c)$.
- **Implicit schemes** (Preissmann four-point box scheme with weighting $\theta \approx 0.6$–1.0) – unconditionally stable, used in HEC-RAS, MIKE 11, SOBEK.
- **Finite-volume / Godunov schemes** with Riemann solvers – best for shocks, bores and dam-break.

2.7 Characteristic form (derivation outline)

Multiply the continuity equation (written with y) by a factor λ and add to the momentum equation, then choose λ such that the combination can be expressed as total derivatives. For a rectangular channel with $c = \sqrt{(gy)}$:

$$\mathbf{C^+: d(V + 2c)/dt = g(S_0 - S_f) \text{ along } dx/dt = V + c}$$

$$\mathbf{C^-: d(V - 2c)/dt = g(S_0 - S_f) \text{ along } dx/dt = V - c}$$

The quantities V ± 2c are called **Riemann invariants** (for a frictionless horizontal channel they stay constant along the characteristics). For $S_0 = S_f = 0$:

V + 2c = constant along C^+; V − 2c = constant along C^-.

These are the basis of the simple-wave solutions for dam-break and negative surges (Sections 4–5).

✎ **Exam Question:** Derive the continuity and dynamic equations for unsteady flow in open channels. List the assumptions. (10 marks)

3. UNIFORMLY PROGRESSIVE WAVE

3.1 Definition

A **uniformly progressive wave** (also called a **monoclinal wave** or **steady wave**) is a wave that moves along the channel with constant velocity V_w and without change of shape. Observed from a reference frame travelling with the wave, the flow appears steady, so the problem reduces to a steady non-uniform flow analysis.

Examples: flood wave in a long channel; a surge in which depth difference is constant; wave maintained by a constant inflow.

3.2 Transformation to steady flow

Let $\xi = x - V_w t$. For a wave of constant shape, $y = y(\xi)$, $Q = Q(\xi)$. Then $\partial/\partial t = -V_w \, d/d\xi$ and $\partial/\partial x = d/d\xi$. Substituting in the continuity equation $\partial Q/\partial x + \partial A/\partial t = 0$:

$dQ/d\xi - V_w \, dA/d\xi = 0 \rightarrow$ **$Q - V_w A =$ constant**.

That is, the discharge relative to the moving frame, $Q_r = (V - V_w)A$, is the same at every section.

3.3 Wave celerity (Kleitz–Seddon law)

Applying the constant-shape condition at two points on the wave (e.g. far upstream and far downstream where uniform flow exists with (Q_1, A_1) and (Q_2, A_2)):

$$\mathbf{V_w = (Q_2 - Q_1)/(A_2 - A_1) \rightarrow dQ/dA = (1/T)\, dQ/dy}$$

This is the **Kleitz–Seddon law**: the celerity of a flood wave equals the slope of the discharge rating curve divided by the top width. If the rating curve is the Manning relation $Q = (1/n)AR^{2/3}S_0^{1/2}$, then (wide channel, $R \approx y$)

$$\mathbf{V_w = (5/3)\ V\ (wide\ rectangular)\ ;\ (13/9)\ V\ (wide\ parabolic)\ ;\ (4/3)\ V\ (triangular)\ [Manning]}$$

With **Chezy** friction a wide rectangular channel gives $V_w = (3/2)V$. So a flood wave moves faster than the water particles – the wave transmits "information" or kinematic effect and the water just passes through.

★ **Key Point:** Relationship between flow velocity and kinematic wave speed: $c_k = dQ/dA = \beta_k V$, where $\beta_k = 1.3–1.7$ (Manning). Typical flood waves travel at 1–3 m/s in lowland rivers.

3.4 Dynamic uniformly progressive wave

Using the dynamic equation with Chezy friction for a wave with $\partial y/\partial \xi \neq 0$, the water surface profile of the wave is given by the integration of

$$\mathbf{dy/d\xi = (S_0 - S_f) \,/\, (1 - (V - V_w)^2/(g\, y))}$$

This looks like the GVF equation with the velocity replaced by the **relative velocity**. A smooth monoclinal wave exists only if the relative Froude number $(V - V_w)/\sqrt{(gy)}$ at the wave is < 1 at the upstream end of the wave; if V_w is such that Fr_r exceeds unity, the front steepens into a bore (hydraulic jump that moves).

3.5 Condition for stability (Vedernikov number)

Roll waves (trains of surges) develop on steep, smooth channels (chutes, spillways) when the Froude number exceeds a limit given by the **Vedernikov–Powell criterion**: for a wide channel, Fr > 2 with Chezy friction and Fr > 1.5 with Manning friction. Roll waves are periodic uniformly progressive waves with hydraulic jumps; they can cause overtopping of chute walls.

3.6 Applications

- Flood routing by kinematic wave in steep catchments and overland flow.
- Estimating flood travel time $T = L/V_w$ in a river reach.
- Irrigation-canal operation: gate-change waves travel with celerity $V \pm c$; control by downstream-gate operation.

✎ **Exam Question:** A flood wave moves in a wide rectangular channel with n = 0.03, S_0 = 0.0004. Normal depth is 2.0 m. Find wave speed. *Ans: $V = (1/n)\, y^{2/3}\, S^{1/2} = (1/0.03)(1.587)(0.02) = 1.058$ m/s; $V_w = 5/3 \times 1.058 = 1.76$ m/s.*

4. RAPIDLY VARIED UNSTEADY FLOW – SURGES

4.1 Definition

A **surge (or wave) of finite height** is a moving hydraulic jump produced when the flow is suddenly changed at a section (gate closing/opening, turbine rejection, tidal inflow). It is a rapidly varied unsteady flow because the water surface rises or falls abruptly at the wave front; hydrostatic assumption is not valid in the front.

- **Positive surge:** the depth increases across the wave ($y_2 > y_1$).
- **Negative surge:** the depth decreases across the wave ($y_2 < y_1$).

Further, according to the direction of travel of the wave front: **upstream-moving** (due to downstream gate closing or inflow reduction/increase upstream) and **downstream-moving** (due to gate opening upstream, or sudden increases in inflow at upstream end).

Cause	Surge type	Direction of travel
Sudden closure of downstream gate	Positive surge	Upstream
Sudden partial closure of upstream gate	Negative surge	Downstream
Sudden opening of upstream gate / increased inflow	Positive surge	Downstream
Sudden opening of downstream gate	Negative surge	Upstream

4.2 Positive surge – basic equations

Consider a surge moving upstream into a channel with steady flow V_1, depth y_1. Behind the surge the flow has V_2 and depth y_2. The surge wave speed (celerity) relative to the bank is V_w. A moving observer rides with the surge so the flow looks steady.

Relative velocities: upstream of surge: $(V_1 + V_w)$, downstream: $(V_2 + V_w)$.

Continuity: $y_1(V_1 + V_w) = y_2(V_2 + V_w)$ (1)

Momentum (per unit width):

$(\gamma/2)(y_1^2 - y_2^2) = \rho\, y_1(V_1 + V_w)[(V_2 + V_w) - (V_1 + V_w)]$ (2)

Eliminating V_2 between (1) and (2):

$$\mathbf{V_w = -V_1 + \sqrt{(\, g\, y_2\, (y_1 + y_2) / (2\, y_1)\,)}}$$

Equivalent forms: $\mathbf{V_w + V_1 = \sqrt{(g\, y_1)}\ \sqrt{[\,½\, (y_2/y_1)(1 + y_2/y_1)\,]}}$. For a still-water upstream ($V_1 = 0$) the surge speed is

$$\mathbf{V_w = \sqrt{(\, g\, y_2(y_1 + y_2)/(2y_1)\,)} \geq \sqrt{(g\, y_1)}}$$

which equals the small-wave celerity when $y_2 \to y_1$. So the larger the surge height, the faster the surge. From continuity (1) the velocity behind the surge is

$$\mathbf{V_2 = y_1(V_1 + V_w)/y_2 - V_w}$$

For still water ahead ($V_1 = 0$) this gives $V_2 = -V_w(y_2 - y_1)/y_2$: the negative sign shows that the water behind a tidal bore moves in the same direction as the bore (upstream).

4.3 Energy loss across the surge

$$h_L = (y_2 - y_1)^3 / (4\, y_1\, y_2)$$

Same expression as the stationary hydraulic jump, since in the moving frame it is a steady jump. The Froude number of the surge: $Fr_s = (V_1 + V_w)/\sqrt{(g\, y_1)}$; if $Fr_s < 1.7$ the front is **undular** (a train of waves following – "undular bore"), $Fr_s > 1.7$ turbulent with white-water roller (e.g., Severn bore, Qiantang bore).

4.4 Worked example – positive surge

A canal flows at $V_1 = 1.0$ m/s with $y_1 = 2.0$ m. A downstream gate is suddenly partly closed and the depth behind the surge becomes $y_2 = 2.4$ m. Find the surge speed and the velocity behind it.

- $V_w = -1.0 + \sqrt{(9.81 \times 2.4 \times 4.4/(2 \times 2.0))} = -1.0 + \sqrt{25.9} = -1.0 + 5.09 =$ **4.09 m/s** (moving upstream relative to the bank).
- Continuity: $V_2 = y_1(V_1 + V_w)/y_2 - V_w = 2.0 \times 5.09/2.4 - 4.09 = 4.24 - 4.09 =$ **0.15 m/s** (downstream). The closure has cut the velocity from 1.0 to 0.15 m/s.
- Energy loss: $h_L = (0.4)^3/(4 \times 2.0 \times 2.4) = 0.0033$ m ($Fr_s = 5.09/4.43 = 1.15$, so the front is undular).

★ **Key Point:** Always draw a diagram with arrows and define signs before applying the continuity and momentum equations. Most mistakes in surge problems are sign errors.

5. NEGATIVE SURGES AND SIMPLE WAVES

5.1 Negative surge

In a negative surge the depth falls across the wave; unlike the positive surge, **there is no stable abrupt front** – energy would have to increase across a "negative jump", which is impossible. Hence a negative surge is a **smooth, gradually flattening wave**, i.e. a depression wave in which each depth travels with its own celerity (a rarefaction fan). The wave flattens as it travels because the deeper water at its leading edge has a larger celerity than the shallower water behind it.

5.2 Analysis using characteristics

Take a frictionless horizontal rectangular channel, initial V_1, y_1 and a flow reduction or lowered depth created at one end. The disturbance is a **simple wave**: all characteristics of one family (here C^+) start in the region of uniform flow, so the Riemann invariant carried by that family has the same value everywhere in the wave.

For a negative wave travelling upstream (into region 1 with y_1, V_1), the invariant along C^+ is constant:

$$\mathbf{V + 2\sqrt{(g\,y)} = V_1 + 2\sqrt{(g\,y_1)} = constant}$$

So the velocity at any depth y in the wave is $\mathbf{V = V_1 + 2(\sqrt{(g\,y_1)} - \sqrt{(g\,y)})}$ ($y < y_1$, so velocity rises as depth falls). The C^- characteristics are straight lines with slope $\mathbf{dx/dt = V - \sqrt{(gy)} = V_1 + 2\sqrt{(g\,y_1)} - 3\sqrt{(gy)}}$. The leading edge (depth y_1) travels at $V_1 - \sqrt{(g\,y_1)}$ (upstream); a lower depth travels more slowly, so the wave spreads out as it advances.

5.3 Comparison

Feature	Positive surge	Negative surge
Depth change	Rise ($y_2 > y_1$)	Fall ($y_2 < y_1$)
Shape of front	Steep, abrupt (moving jump)	Gradual, flattening (spreads)
Energy	Lost across the front ($h_L > 0$)	Conserved (isentropic)
Method	Continuity + momentum across jump	Method of characteristics (Riemann invariants)
Celerity	Same everywhere at the front	Varies with depth, $V \pm \sqrt{(gy)}$
Example	Tidal bore; gate closing	Gate opening downstream; sudden pump shutoff upstream

5.4 Applications and engineering significance

- Design of **freeboard** and canal lining: positive surges following turbine rejection in power canals may overtop banks.
- **Power canals/penstock forebays:** upstream surge, negative surge on load demand increase.
- **Irrigation canal gates:** operation rules to limit surge height (≤ 10–20 % of normal depth).
- **Tidal bores:** navigation hazard; used for surfing; tidal power; sediment resuspension.
- **Pumping station start and stop;** emergency closure gates.

✎ **Exam Question:** Distinguish between positive and negative surges. Derive the expression for celerity of a positive surge in a rectangular channel. (10 marks)

6. DAM BREAK PROBLEM

6.1 Introduction

A **dam break** is a catastrophic failure of a dam which suddenly releases the stored water as a flood wave, travelling downstream. Examples: Teton Dam (1976), Banqiao (1975), Machhu-II (Morbi, India, 1979), Malpasset (France, 1959), St Francis (USA, 1928). Dam-break analysis serves **emergency action plans (EAPs)**, flood inundation mapping and risk assessment.

Failure modes: overtopping, piping/internal erosion, foundation failure, earthquake, landslide-generated waves.

Types: **instantaneous failure** (concrete arch or gravity dams – complete, sudden) and **progressive failure** (earth/rockfill – a breach grows over minutes to hours).

6.2 Ritter's solution – instantaneous total collapse on a dry bed

Assumptions: horizontal, frictionless, wide rectangular channel; dam removed instantly at t = 0; initial still water depth y_0 upstream; dry bed downstream. Hydrostatic pressure; 1-D flow.

Initial: V = 0, y = y_0 for x < 0; y = 0 for x > 0 (dam at x = 0). Reservoir celerity $c_0 = \sqrt{(g\, y_0)}$.

Characteristics: the **negative wave** moves upstream into the reservoir with speed $-c_0$ (i.e. the leading edge of the disturbance is at $x = -c_0\, t$). Since the region upstream is initially at rest, the Riemann invariant V + 2c along C^+ carries the value $2c_0$:

$$\mathbf{V + 2c = 2c_0 \rightarrow V = 2(c_0 - c)}$$

and the C^- characteristics are straight lines: $dx/dt = V - c = 2c_0 - 3c$, so $x/t = 2c_0 - 3c$. Solve for c and V:

$$\mathbf{c = (2c_0 - x/t)/3 \rightarrow y(x,t) = (1/9g)(2c_0 - x/t)^2}$$

$$\mathbf{V(x,t) = (2/3)(c_0 + x/t)}$$

valid for $\mathbf{-c_0 t \le x \le 2c_0 t}$. Wave front on the dry bed travels at $\mathbf{2c_0 = 2\sqrt{(g\, y_0)}}$; the water surface profile is a **parabola**.

Location	Depth	Velocity
$x = -c_0 t$ (upstream limit of disturbance)	y_0	0
x = 0 (dam site)	$\mathbf{4y_0/9} = 0.444 y_0$	$\mathbf{(2/3)c_0} = (2/3)\sqrt{(g y_0)}$
$x = 2c_0 t$ (front)	0	$2c_0$

Discharge at the dam site: $\mathbf{q = y\, V = (8/27)\, y_0\, \sqrt{(g\, y_0)}}$ per unit width, and it is **constant in time** (until the negative wave reaches the far end of the reservoir). Fr at x = 0 is exactly 1 – critical flow at the dam site.

★ **Key Point:** Ritter's solution overestimates the speed of the front since friction is neglected; with friction, the front is blunt, thick and travels slower than $2c_0$ (Dressler 1952).

6.3 Worked example

A dam holding y_0 = 40 m of water fails suddenly; the tailwater is dry. $c_0 = \sqrt{(9.81 \times 40)} = 19.8$ m/s. Depth at dam site = 17.8 m; velocity = 13.2 m/s; discharge per metre width = 235 m^2/s; front speed = 39.6 m/s (142 km/h, theoretical). After 5 minutes the front would be 11.9 km downstream (without friction).

7. DAM BREAK – WET BED (STOKER'S SOLUTION) AND PARTIAL BREACH

7.1 Dam break on a wet bed (tailwater depth y_1)

When there is initial water of depth y_1 ($< y_0$) downstream, a **positive surge** (bore) moves downstream with speed V_w and a **negative wave** moves upstream into the reservoir. Between them is a zone of constant depth y_m and velocity V_m.

Negative wave (Riemann invariant from reservoir): $V_m = 2(c_0 - c_m)$, $c_m = \sqrt{(g\, y_m)}$.

Positive surge (jump conditions): $V_w = V_m\, y_m/(y_m - y_1)$ and $V_w^2 = g\, y_m(y_m + y_1)/(2y_1)$.

Combining these gives a single non-linear equation for y_m (solve by Newton–Raphson):

$$\mathbf{2(c_0 - c_m) = (y_m - y_1)\,\sqrt{[\, g\,(y_m + y_1)/(2\, y_m\, y_1)\,]}}$$

Example: $y_0 = 10$ m, $y_1 = 1$ m. Here $c_0 = 9.9$ m/s; iteration gives $y_m \approx 3.96$ m, $V_m \approx 7.35$ m/s and bore speed $V_w \approx 9.8$ m/s. Ratio $y_1/y_0 \to 0$ gives the dry-bed limit; ratio $\to 1$ gives a weak wave.

7.2 Effect of friction and channel slope

Real dam-break waves attenuate (peak discharge decreases and wave flattens) as they move downstream due to resistance, storage in the valley (floodplain), channel widening and loss of volume. For analysing real valleys:

- Use the full Saint-Venant equations with implicit or Godunov schemes.
- Use dimensionless parameters: $Q_* = Q/Q_{max}$, distance $X_* = x/L$.
- Use empirical formulas for peak outflow from breach (Froehlich 1995): $Q_p = 0.607\, V_{res}^{0.295}\, h^{1.24}$ (SI), V_{res} = reservoir volume (m^3), h = height of water above the breach bottom (m).

7.3 Peak flow estimation

- **Instantaneous failure** of a rectangular section: $Q_{max} = (8/27)\, b\, \sqrt{g}\, y_0^{3/2}$ (Ritter).
- **Linear breach with time:** the breach width grows linearly; the peak flow from the broad-crested weir equation $Q = C\, b\, y^{3/2}$.
- **Peak attenuation:** the peak falls as the wave moves down the valley because of valley storage and friction; it is estimated by storage routing (e.g. Wetmore–Fread) or numerical models.

7.4 Dam-break analysis workflow

- 1. Collect reservoir data: volume–elevation curve, dam geometry, breach scenario (sunny-day, flood overtopping).
- 2. Estimate breach parameters: width, side slopes, formation time (empirical equations – MacDonald & Langridge-Monopolis, Froehlich, Xu & Zhang).
- 3. Compute outflow hydrograph through the breach (weir-flow, reservoir routing).
- 4. Route through downstream valley (HEC-RAS 1D/2D, MIKE 11, DAMBRK, BREACH, FLO-2D).
- 5. Map flood extent, depth, velocity, arrival time and hazard classification.
- 6. Prepare **Emergency Action Plan**: inundation maps, warning, evacuation, communication.

7.5 Dam-break and surge – key comparisons

Aspect	Dry-bed (Ritter)	Wet-bed (Stoker)
Front type	Smooth, tip with zero depth	Steep bore (jump) of finite height
Front speed	$2c_0$	$V_w < 2c_0$, from jump equation
Depth at dam site	$4y_0/9$ (constant)	$4y_0/9$ if $y_m < 4y_0/9$, otherwise y_m
Method	Characteristics (simple wave)	Characteristics + jump conditions

✎ **Exam Question:** Derive Ritter's solution for the instantaneous dam-break problem on a dry horizontal bed. Show that the depth at the dam site is 4/9 y_0. (10 marks)

8. NUMERICAL METHODS AND QUICK REVISION

8.1 Numerical solution of the Saint-Venant equations

Explicit schemes: simple, but stable only if Courant number Cr = (|V| + c)Δt/Δx ≤ 1. Lax diffusive scheme, MacCormack predictor–corrector scheme (second-order accurate, good for shocks with artificial viscosity).

Implicit schemes: Preissmann box scheme; weighted averages of variables at n and n+1 time levels give a banded (block-tridiagonal) system solved by the double-sweep method. Stable for θ ≥ 0.5; Δt can be large (but accuracy reduces).

Finite-volume / Godunov: solves Riemann problem at each cell interface (Roe, HLL, HLLC solvers); handles discontinuities and wet–dry fronts. Used in modern dam-break models (HEC-RAS 2D, TELEMAC, LISFLOOD).

8.2 Boundary and initial conditions

- **Initial:** steady-flow profile (backwater calculation) for all x.
- **Upstream boundary:** inflow hydrograph Q(t) (flood) or stage hydrograph.
- **Downstream boundary:** rating curve Q(y), stage hydrograph (tide) or normal depth.
- **Internal:** gates, weirs, bridges, culverts – given by head–discharge relations.

8.3 Flood routing methods in brief

Method	Equation	Parameters
Kinematic wave	$\partial A/\partial t + c_k\, \partial A/\partial x = 0$	$c_k = dQ/dA$
Muskingum	$S = K[XI + (1-X)O]$	K (travel time), X (0–0.5)
Diffusion wave	$\partial Q/\partial t + c\, \partial Q/\partial x = D\, \partial^2 Q/\partial x^2$	Celerity and diffusivity $D = Q/(2TS_0)$
Dynamic wave	Full Saint-Venant	Requires numerical scheme

8.4 Formulae sheet

	Formula		
	$c = \sqrt{(gy)}$		
	$\partial A/\partial t + \partial Q/\partial x = 0$		
	$S_f = S_0 - \partial y/\partial x - (V/g)\partial V/\partial x - (1/g)\partial V/\partial t$		
	$V_w = dQ/dA = (1/T)dQ/dy$		
	$V_w = -V_1 + \sqrt{(gy_2(y_1 + y_2)/2y_1)}$		
	$h_L = (y_2 - y_1)^3/(4y_1y_2)$		
	$y_{dam} = 4y_0/9$; front speed $= 2\sqrt{(gy_0)}$		
	Δt ≤ Δx/(	V	+ c)

8.5 Important short questions

- What are the assumptions of the Saint-Venant equations?
- Define uniformly progressive wave. State the Kleitz–Seddon law.
- Why does a negative surge not form an abrupt front?
- What are Riemann invariants?
- State the Courant–Friedrichs–Lewy (CFL) condition.
- Differentiate kinematic, diffusion and dynamic wave models.

- List causes of dam failure and two methods of analysis.